
Training-Free Constraint Projection for Instruction-Faithful 3D Indoor Scene Layouts

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Abstract

Instruction-guided 3D scene generators can produce plausible rooms while leaving explicit spatial constraints unsatisfied in the final continuous layout. We introduce Relation-Aware Constraint Projection, a training-free post-generation framework that extracts or consumes spatial constraints, verifies them on object geometry, and projects a generated layout toward constraint satisfaction using bounded floor-plane edits and safety gates. On the original InstructScene validation splits, the collision-gated Floor-Prior variant improves realized relation accuracy by 7.8, 8.8, and 7.8 absolute points for bedrooms, living rooms, and dining rooms, respectively, while reducing mesh-collision pair rates. On 339 CommonScenes outputs spanning four room types, the same generic repair principle yields smaller but consistent relation-score gains of 0.16–0.54 points, reduces collision penalties in every room type, and changes 1.4–4.1% of boxes. A strict one-factor study on 162 InstructScene bedrooms further reduces ground-plane overlap by 42.7% with 0.075 m mean movement and no scene-level degradation in scene-graph/layout consistency. These results support final-layout constraint projection as a low-perturbation control layer for pretrained 3D scene generators, while also exposing limits when the baseline already nearly satisfies the chosen constraints.

1 Introduction

Text-conditioned 3D indoor scene generation is useful for embodied AI, simulation, game production, and interactive design. In these settings, spatial language is not decorative. If a prompt asks for a nightstand behind a bed or a coffee table to the right of a side table, the generated layout must realize that relation in geometry. Recent indoor scene generators synthesize plausible object arrangements from scene categories, floor plans, scene graphs, or text [6–9]. However, a symbolic relation that appears in an intermediate graph does not guarantee that the final continuous layout satisfies the requested relation.

This paper studies final-layout constraint violations in InstructScene [6], an instruction-driven 3D indoor scene synthesis pipeline with a semantic graph prior, and in CommonScenes [11], a scene-graph diffusion model. We avoid attributing every violation to one internal decoding stage: the available evidence establishes a mismatch between requested or generated constraints and final geometry, but does not isolate a single causal source. Instead, we ask whether a lightweight verifier-repair layer can improve the final layout while keeping pretrained generators fixed.

We propose Relation-Aware Constraint Projection, a plug-in final-layout control layer. Given a generated layout and either text-derived or graph-derived relations, the module matches constraints to generated objects, verifies whether floor-plane geometry satisfies them, and repairs translations when a relation or overlap constraint is violated. The final design edits only the horizontal floor plane, preserves object height, limits displacement, and can fall back to the baseline when mesh collision or scene-graph/layout consistency would worsen. These gates matter because unconstrained geometric repair can improve a target metric by moving furniture implausibly far or introducing new collisions.

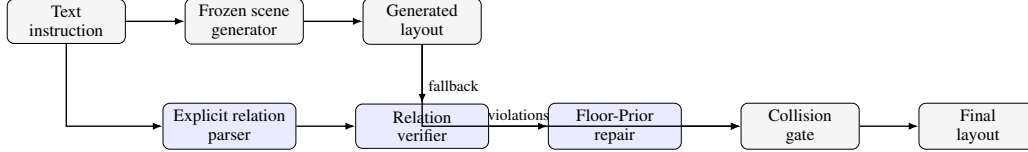


Figure 1: **Relation-Aware Constraint Projection overview.** We keep the pretrained generator fixed, obtain constraints from text or a generated scene graph, verify the final continuous layout, and project violated relations in the floor plane while preserving object height. Safety gates reject candidates that worsen mesh collision or the guarded scene-graph/layout consistency metric.

Our experiments cover the original InstructScene bedroom, living-room, and dining-room validation splits, a four-room CommonScenes study, and a strict layout-only ablation. On InstructScene, collision-gated relation gains remain positive for all room types and mesh-pair rates decrease. On CommonScenes, where generic relation satisfaction already exceeds 0.95, the method makes correspondingly small edits and produces small but consistent generic-score gains while the official CommonScenes score slightly decreases. We report this negative result because it distinguishes improvement on our generic geometric verifier from improvement on an independent benchmark metric. The strict bedroom ablation additionally shows that a scene-graph consistency guard can reduce overlap without sacrificing its guarded metric.

Our contribution is not a claim of first relation-aware 3D scene generation, nor a claim that a post-generation layer replaces stronger full generators. Strong related systems use scene graphs, semantic dependency graphs, language models, and optimization to reason about spatial constraints [1, 2, 5, 10, 11]. Instead, this paper makes a narrower and reproducible claim:

- We formulate post-generation 3D layout repair as training-free constraint projection with explicit verification, bounded edits, and fallbacks.
- We evaluate two complementary safety gates—mesh-collision gating and scene-graph consistency guarding—and quantify their accuracy–movement–collision trade-offs.
- We test the repair principle on both InstructScene and CommonScenes, report consistent in-domain gains, and retain the independent CommonScenes score decrease as a claim boundary rather than hiding it.

2 Related Work

Indoor scene synthesis. 3D-FRONT [4] and 3D-FUTURE [3] provide large-scale room layouts and textured furniture assets that underpin much recent indoor scene synthesis work. ATISS [7] models indoor scenes as unordered sets generated autoregressively. SceneFormer [9] uses transformers to generate indoor scenes conditioned on room layouts. DiffuScene [8] uses denoising diffusion over unordered object attributes. These methods focus on plausible and diverse indoor layouts; our work focuses on whether explicit natural-language relations are realized in the final layout of an instruction-conditioned pipeline.

Text and graph conditioned scene generation. InstructScene [6] is our direct baseline because it synthesizes 3D indoor scenes from natural language instructions with a semantic graph prior. CommonScenes [11] converts scene graphs into commonsense 3D scenes with scene graph diffusion. LayoutGPT [2] uses large language models as visual planners for layouts, including spatial and numerical constraints. Holodeck [10] uses language models and 3D assets to generate embodied AI environments, including spatial constraints for object placement. These works show that language and structured relations are useful interfaces for scene generation. Our contribution is complementary: we evaluate and repair the final realized relation after a pretrained instruction-to-scene pipeline has produced a layout.

Strong relation-aware systems. ReSpace [1] is a close text-driven indoor scene synthesis and editing framework based on structured scene representations and a specialized language model. SDGScenes [5] uses a semantic dependency graph, VLM-derived commonsense constraints, and

nonlinear constrained optimization to align scenes with user intent. These systems overlap with the broad goal of relation- and intent-aware 3D generation. We therefore do not claim to outperform them or to be the first relation-aware method. Our controlled comparisons use frozen InstructScene and CommonScenes outputs; ReSpace and SDGScenes require a shared prompt, asset, and evaluation protocol before direct numerical claims are appropriate. Put differently, we do not replace strong generators; we study a plug-in final-layout relation grounding layer for existing generators.

3 Method

3.1 Problem Setup

Let an instruction-conditioned generator produce a scene layout

$$S = \{(c_i, t_i, s_i, r_i, m_i)\}_{i=1}^N,$$

where object i has category c_i , translation $t_i \in \mathbb{R}^3$, size s_i , yaw rotation r_i , and optionally a retrieved mesh identifier m_i . The instruction text x contains a set of explicit spatial relations

$$R(x) = \{(c_a, p, c_b)\},$$

where predicate p may be left-of, right-of, in-front-of, behind, near, or far. The goal is to produce a repaired layout S' that improves the realized satisfaction of $R(x)$ while preserving scene plausibility.

3.2 Explicit Relation Extraction and Matching

We use a deliberately simple rule-based parser to extract explicit spatial phrases. This choice makes the method transparent and keeps the benchmark unchanged. The parser output is noisy, but it is sufficient to drive a verifier: the deployable parsed-relation repair captures most of the oracle-repair gain in earlier diagnostics. The parser is not intended to solve implicit commonsense reasoning; prompts that require latent affordance or support reasoning are outside the current claim.

For each parsed triple, the verifier finds generated objects whose class labels match the subject and object categories. When multiple instances match, the verifier selects the nearest pair in the floor plane. All distance and direction checks use the x/z plane; the y axis is treated as vertical height.

3.3 Relation Verification

For a candidate object pair (i, j) and predicate p , the verifier evaluates whether the floor-plane displacement $\Delta_{ij} = (t_i^x - t_j^x, t_i^z - t_j^z)$ satisfies the predicate. Directional predicates check the dominant signed displacement along the relevant axis; distance predicates compare floor-plane distance against close or far thresholds. We report exact selected-relation accuracy, counting a relation correct only when the target relation appears in the realized relation set.

3.4 Direct Height-Preserving Repair

The direct repair baseline moves an object in the floor plane to satisfy a violated relation. It preserves the original height coordinate:

$$t'_i = (t_i^x + \delta_x, t_i^y, t_i^z + \delta_z).$$

This repair gives large relation gains but may move objects substantially. In our experiments, direct repair increases relation accuracy by up to 19.7 points, but average repair movement is about two meters and mesh collision can increase slightly. We therefore use direct height-preserving repair as an internal ablation, not as the main method.

3.5 Floor-Prior Repair

The Floor-Prior repair constrains relation repair with scene priors. For a violated relation, the method proposes candidate translations with interpolation strengths $\alpha \in \{0.25, 0.5, 0.75, 1.0\}$, clamps movement by a maximum radius, and scores candidates by relation satisfaction and coarse overlap penalties. The chosen candidate must improve the relation while keeping the motion bounded. In all variants, the vertical coordinate remains unchanged.

Algorithm 1: Floor-Prior relation repair

Input: Layout S , parsed relations $R(x)$, thresholds $d_{\text{close}}, d_{\text{far}}$, max movement ρ , passes K

Output: Repaired layout S'

$S' \leftarrow S$

for $k = 1$ **to** K **do**

foreach $(c_a, p, c_b) \in R(x)$ **do**

$(i, j) \leftarrow$ nearest matching object pair in the x/z plane

if p is already satisfied by (i, j) **then**

 continue

end

$\mathcal{C} \leftarrow$ bounded floor-plane candidate translations for object i

 Score each $c \in \mathcal{C}$ by relation satisfaction and overlap penalty

 Apply the best candidate, preserving t_i^y

end

end

return S'

3.6 Scene-Graph-Guarded Overlap Projection

When a generator exposes its predicted scene graph, we also instantiate constraint projection as overlap reduction with a scene-graph consistency guard. The repair visits intersecting oriented bounding-box footprints, proposes bounded translations that push a pair apart in the floor plane, and accepts a proposal only when total overlap strictly decreases and scene-graph/layout consistency does not fall below the baseline for that scene. A scene-level guard restores the original layout if the final overlap exceeds the baseline. This variant measures consistency with the generator’s own graph; it is a self-consistency diagnostic, not accuracy against human or ground-truth relations.

3.7 Collision-Gated Selection

For mesh-aware evaluation, we load the retrieved 3D-FUTURE meshes and compute FCL mesh-collision pair counts using a trimesh collision manager. The collision-gated variant keeps the repaired scene only when the repaired mesh collision pair count does not exceed the baseline count; otherwise it falls back to the baseline layout for that scene. This gate is a verifier-selection layer and should be described explicitly.

4 Experiments

4.1 Experimental Setup

We evaluate on the existing InstructScene validation splits for bedrooms, living rooms, and dining rooms. We use the official pretrained SG diffusion VQ object-feature checkpoints: bedroom epoch 1999, living-room epoch 1459, and dining-room epoch 1239. We additionally evaluate frozen CommonScenes epoch-195 outputs for bedrooms, living rooms, dining rooms, and libraries. We do not create a new benchmark or add new annotations. Metrics include realized relation accuracy, average repair movement in the floor plane, mesh collision pair rate, generic relation satisfaction, official CommonScenes constraint score, ground-plane overlap, scene-graph/layout consistency, and height-change diagnostics. Visual diagnostics use the same before/after renderer, SSIM, PSNR, and scene-level GLB Chamfer distance.

4.2 Main Results

Table 1 compares the original InstructScene layout with Floor-Prior repair. The main repair improves relation realization in all three room categories, with gains from 10.2 to 14.3 absolute points. The non-gated mesh pair delta is small but positive, so mesh-improvement claims should use the gated results in Table 3.

Table 1: Main Floor-Prior relation results. Accuracy is exact realized selected-relation accuracy. Movement is the average floor-plane movement for repaired pairs.

Room	Baseline	Floor-Prior	Gain	Move	Mesh Δ
Bedroom	0.7388	0.8408	+0.1020	0.7837	+0.0029
Dining room	0.5948	0.7286	+0.1338	0.7315	+0.0008
Living room	0.5510	0.6939	+0.1429	0.8808	+0.0008

Table 2: Direct height-preserving repair. Direct repair is a strong internal baseline but moves objects much more than Floor-Prior repair.

Room	Baseline	Direct	Gain	Move	Mesh Δ
Bedroom	0.7388	0.8735	+0.1347	2.2101	+0.0034
Dining room	0.5948	0.7844	+0.1896	1.9694	+0.0019
Living room	0.5510	0.7483	+0.1973	2.2834	+0.0004

4.3 Direct Repair Is Strong but Too Aggressive

Table 2 shows that direct height-preserving repair achieves larger relation gains than Floor-Prior repair, but with substantially larger movements. This result motivates the floor-prior design: a paper-facing method should not win relation accuracy merely by pushing furniture far across the room.

4.4 Collision-Gated Results

Table 3 reports the conservative collision-gated Floor-Prior variant. The gate reduces relation gains relative to ungated repair because it falls back on scenes where mesh collision would increase. Nevertheless, gated relation gains remain positive in every room, and gated mesh pair rates fall below baseline. We use only this gated variant for mesh-validity claims.

4.5 Multi-Seed Stability

Table 4 summarizes three-seed results for the main Floor-Prior configuration. Ungated relation gains are stable and positive across all rooms. After collision gating, relation gains remain positive, and the mean mesh delta is negative for every room.

4.6 Cross-Generator Evaluation on CommonScenes

Table 5 evaluates the generic movement/collision-gated repair on 339 frozen CommonScenes outputs. The baseline already achieves high generic relation satisfaction, leaving little headroom. The repair nevertheless improves the generic score and reduces collision penalty for every room type while editing only 1.4–4.1% of boxes. The CommonScenes official score is an independent diagnostic and decreases on the bedroom split from 0.9732 to 0.9696; therefore, we do not claim improvement on the official CommonScenes metric.

4.7 Strict One-Factor Safety Ablation

We run a strict one-factor study on 162 InstructScene bedroom layouts using the scene-graph-guarded overlap projection. The recommended setting uses a 0.05 m step, 0.25 m displacement cap, and 50 iterations. It reduces the aggregate oriented-footprint IoU sum from 0.2245 to 0.1287 (42.7%), increases scene-graph/layout consistency from 0.9095 to 0.9134, and moves objects by 0.0753 m on average. Across 1,296 repairs from eight evaluated configurations, no run fails, no scene has worse overlap, and no scene has worse guarded consistency. Iteration limits of 25, 50, and 75 produce identical outputs, indicating convergence by 25 iterations under the tested setting.

Table 3: Collision-gated Floor-Prior results. The gate keeps repaired layouts only when FCL mesh collision pairs do not increase.

Room	Base	Repair	Gated	Gated gain	Base mesh	Gated mesh	Fallback
Bedroom	0.7388	0.8408	0.8163	+0.0776	0.0702	0.0683	6
Dining room	0.5948	0.7286	0.6729	+0.0781	0.0342	0.0336	15
Living room	0.5510	0.6939	0.6395	+0.0884	0.0313	0.0310	14

Table 4: Three-seed stability. Standard deviations are population standard deviations over three seeds.

Room	Mean gain	Std	Mean gated gain	Gated std	Mean gated mesh Δ
Bedroom	+0.1061	0.0058	+0.0803	0.0069	-0.0020
Dining room	+0.1462	0.0093	+0.0867	0.0098	-0.0008
Living room	+0.1270	0.0125	+0.0703	0.0153	-0.0003

4.8 Movement Ablation

A conservative movement radius of 0.8 meters reduces average movement to 0.38–0.42 meters but also reduces relation gains to 5.7–7.8 points. The main 1.8-meter setting gives a better relation-control tradeoff, while the 0.8-meter variant is useful when minimal movement is more important than maximum relation satisfaction.

4.9 Height Preservation and Visual Diagnostics

The height diagnostic reports y -changed rate 0.0 and maximum absolute y delta 0.0 for the evaluated Floor-Prior variants. Thus the final repair does not move furniture vertically. This fixes a practical failure observed in early visualizations where chairs or sofas could appear to float.

We also rendered real 3D-FUTURE mesh comparisons. Table 8 compares the old three-example direct showcase with the new 15-example Floor-Prior showcase selected for relation improvement without mesh-pair increase. The Floor-Prior showcase has higher mean relation gain, lower mesh pair count, slightly positive CLIP delta, and high SSIM. We interpret these diagnostics conservatively: they support that selected examples do not show obvious visual degradation, but they do not prove universal human-perceived visual improvement.

4.10 Baseline Coverage and Claim Boundary

The fully controlled baselines are the frozen generator outputs and internal repair ablations operating on identical scenes. A partial ReSpace reproduction was attempted, but a same-split, same-prompt, same-asset protocol was not completed; SDGScenes likewise has no verified same-protocol runnable path in this repository. We therefore treat ReSpace and SDGScenes as related systems rather than numerical baselines. A random displacement control, a generic nonlinear solver under the same movement budget, an oracle-parser upper bound under the final gated protocol, and a human preference study remain necessary for a strong comparative submission.

5 Discussion

The results support a narrow but useful conclusion: post-generation constraint projection can improve explicit spatial relation realization in an existing instruction-guided scene generator and can produce small, consistent changes on a second generator. The difference between direct repair and safety-aware projection is important. Direct repair shows the upper effect size of geometric correction, but it is too aggressive for a polished method. Bounded repair gives smaller gains with unchanged object heights, movement control, and explicit fallbacks. The CommonScenes result further shows that effect size depends on baseline headroom: a near-saturated generic verifier admits only small improvements.

The results also reveal an evaluation lesson. Reporting only symbolic scene graph correctness can miss failures in the final continuous layout. Conversely, reporting only visual examples can hide

Table 5: CommonScenes cross-room evaluation. Generic Δ is repaired minus baseline generic relation satisfaction. Collision values are penalty before and after repair.

Room	Scenes	Generic Δ	Collision	Edited boxes	Mean move
Bedroom	162	+0.0031	164.01→161.49	1.38%	0.0155 m
Living room	52	+0.0022	54.80→51.84	3.65%	0.0374 m
Dining room	69	+0.0016	103.20→99.13	2.34%	0.0185 m
Library	56	+0.0054	72.03→68.50	4.05%	0.0387 m

Table 6: Strict bedroom one-factor ablation. All settings preserve or improve the guarded scene-graph/layout consistency metric.

Setting	Overlap IoU	SG consistency	Mean move	Max move
Baseline	0.2245	0.9095	0.0000	0.0000
Step 0.025, cap 0.25	0.1347	0.9137	0.0698	0.2500
Step 0.050, cap 0.25	0.1287	0.9134	0.0753	0.2500
Step 0.075, cap 0.15	0.1507	0.9133	0.0566	0.1500
Step 0.075, cap 0.35	0.1216	0.9135	0.0885	0.3474

whether the requested relation is actually satisfied. A relation-aware scene generation benchmark should evaluate instruction parsing, graph-level relation accuracy, final layout relation accuracy, and geometric side effects separately.

6 Limitations

The current method targets explicit or graph-provided spatial relations. It does not handle all commonsense requirements, such as affordance, reachability, support, or task-specific functional arrangements. The rule-based parser has only moderate precision and recall, and the same relation logic participates in both repair and part of the evaluation. The CommonScenes official metric decreases slightly, demonstrating that gains under the generic verifier do not automatically transfer to an independent constraint definition. The collision and scene-graph gates are post-hoc verifier-selection layers, not an end-to-end physically aware generator. The strict ablation is bedroom-only, single-seed, and layout-only; its overlap metric is a 2D oriented-box proxy rather than 3D mesh collision. Our visual diagnostics do not replace a human preference study. Finally, we do not claim superiority over ReSpace, SDGScenes, or other strong related methods without a shared protocol.

7 Conclusion

We presented Relation-Aware Constraint Projection, a training-free verifier-repair layer for pretrained 3D indoor scene generators. The method verifies final layout geometry and applies bounded floor-plane edits that preserve object heights and can be rejected by collision or scene-graph consistency gates. InstructScene experiments show substantial gains in realized explicit relations; CommonScenes experiments show smaller cross-generator gains with low edit rates; and the strict bedroom ablation quantifies the overlap-consistency-movement trade-off. The evidence supports constraint projection as a practical control layer, but independent evaluation and stronger same-protocol baselines remain necessary before claiming state-of-the-art performance.

Reproducibility Notes

The original InstructScene data are archived under `experiment_archive/relation_aware_instructscene_20260530`. The strict ablation tables and reproduction instructions are under `results/bedroom_162_layout_summary`, and the CommonScenes cross-room summaries, bootstrap intervals, and independent-score diagnostics are under `docs/commonsences_experiment_pack_20260713`. Raw datasets and checkpoints are excluded because they are large third-party assets; their expected versions and remote reproduction commands are documented under `server_setup`.

Table 7: Conservative max-move ablation. Smaller movement budgets reduce geometry changes but also reduce relation gains.

Room	Baseline	Max 0.8 repair	Gain	Move
Bedroom	0.7388	0.7959	+0.0571	0.4233
Dining room	0.5948	0.6729	+0.0781	0.3881
Living room	0.5510	0.6156	+0.0646	0.3797

Table 8: Real-mesh visual diagnostics. CLIP delta is after-minus-before text-image similarity; SSIM measures before/after structural similarity.

Showcase	Examples	Relation gain	Mesh-pair Δ	CLIP Δ	SSIM
Direct y-fixed	3	+1.333	0.000	-0.0165	0.8358
Floor-Prior	15	+1.400	-0.267	+0.0027	0.9065

References

- [1] Martin J. J. Bucher and Iro Armeni. Respace: Text-driven 3d scene synthesis and editing with preference alignment, 2025. URL <https://arxiv.org/abs/2506.02459>.
- [2] Weixi Feng, Wanrong Zhu, Tsu-jui Fu, Varun Jampani, Arjun Akula, Xuehai He, Sugato Basu, Xin Eric Wang, and William Yang Wang. Layoutgpt: Compositional visual planning and generation with large language models, 2023. URL <https://arxiv.org/abs/2305.15393>.
- [3] Huan Fu, Rongfei Jia, Lin Gao, Mingming Gong, Binqiang Zhao, Steve Maybank, and Dacheng Tao. 3d-future: 3d furniture shape with texture, 2020. URL <https://arxiv.org/abs/2009.09633>.
- [4] Huan Fu, Bowen Cai, Lin Gao, Ling-Xiao Zhang, Jiaming Wang, Cao Li, Qixun Zeng, Chengyue Sun, Rongfei Jia, Binqiang Zhao, and Hao Zhang. 3d-front: 3d furnished rooms with layouts and semantics. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10933–10942, 2021.
- [5] Yuhan Gao, Xueying Li, Jicong Ao, Chao Yu, Peng Liu, and Chenjia Bai. Sdgsenes: User-intent driven indoor scene generation via semantic dependency graph. *Pattern Recognition*, 179: 113674, 2026. doi: 10.1016/j.patcog.2026.113674. URL <https://doi.org/10.1016/j.patcog.2026.113674>.
- [6] Chenguo Lin and Yadong Mu. Instructscene: Instruction-driven 3d indoor scene synthesis with semantic graph prior, 2024. URL <https://arxiv.org/abs/2402.04717>.
- [7] Despoina Paschalidou, Amlan Kar, Maria Shugrina, Karsten Kreis, Andreas Geiger, and Sanja Fidler. Atiss: Autoregressive transformers for indoor scene synthesis, 2021. URL <https://arxiv.org/abs/2110.03675>.
- [8] Jiapeng Tang, Yinyu Nie, Lev Markhasin, Angela Dai, Justus Thies, and Matthias Nießner. Diffuscene: Denoising diffusion models for generative indoor scene synthesis, 2024. URL <https://arxiv.org/abs/2303.14207>.
- [9] Xinpeng Wang, Chandan Yeshwanth, and Matthias Nießner. Sceneformer: Indoor scene generation with transformers, 2021. URL <https://arxiv.org/abs/2012.09793>.
- [10] Yue Yang, Fan-Yun Sun, Luca Weihs, Eli VanderBilt, Alvaro Herrasti, Winson Han, Jiajun Wu, Nick Haber, Ranjay Krishna, Lingjie Liu, Chris Callison-Burch, Mark Yatskar, Aniruddha Kembhavi, and Christopher Clark. Holodeck: Language guided generation of 3d embodied ai environments, 2024. URL <https://arxiv.org/abs/2312.09067>.
- [11] Guangyao Zhai, Evin Pinar Örnek, Shun-Cheng Wu, Yan Di, Federico Tombari, Nassir Navab, and Benjamin Busam. Commonsences: Generating commonsense 3d indoor scenes with scene graph diffusion, 2023. URL <https://arxiv.org/abs/2305.16283>.

Table 9: Key archived tables used in this draft.

Table	Rows	Columns
<code>floor_prior_results.csv</code>	33	15
<code>floor_prior_gated_results.csv</code>	33	14
<code>floor_prior_mesh_results.csv</code>	33	11
<code>visual_quality_real_mesh.csv</code>	18	31
<code>strong_baseline_feasibility.csv</code>	6	7

A Additional Result Tables

B Submission Checklist Status

This research draft includes claim boundaries, compute and environment notes, limitations, and artifact paths. Before submission, the authors must add a same-budget random or solver baseline, an independent relation evaluator, uncertainty intervals for relation-score gains, a human preference study if making perceptual-quality claims, and the checklist required by the selected venue.